

Practical 2 - Logistic regression

Statistical Models for Discrete Outcomes

London School of Hygiene & Tropical Medicine

Datasets required: `lbw.RData` and `insecticide.RData`

Introduction

In the first part of this session we will use a publicly available dataset on birthweights of babies, taken from the book *Applied Logistic Regression* by David W Hosmer and Stanley Lemeshow. Per the book:

“Data were collected as part of a larger study at Baystate Medical Center in Springfield, Massachusetts. This dataset contains information on 189 births to women seen in the obstetrics clinic. Fifty nine of these births were defined as low birth weight [defined as less than 2500 grams].”

In the second part of the session we will use data from a toxicological experiment. Eight groups of insects were exposed for five hours to gaseous carbon disulphide (CS_2) at different concentrations, with the purpose of investigating how the risk of death depends on the dose received.

Aims

The aims of this session are:

- understand how to interpret results from logistic regression models.
- understand how to compare the fits of logistic regression models.

Part A: Investigation into low birthweights in Massachusetts, USA

We will first investigate factors which may be related to the chance of having a low birthweight baby, using the `lbw` dataset. Begin by loading the data with `load("lbw.RData")` (remember to set your working directory or alternatively add the path/folder to the call to `load()`).

Question 1

Summarise the dataset, checking for missing values. Identify which variable stores our outcome of interest. Also note that the mother's weight is given in lbs, whilst the baby's weight is given in grams.

Question 2

Use the `glm` function to fit a logistic model for the low birthweight variable with the mother's weight at last menstrual period (in lbs) (`lwt`) as the single covariate.

Convert the estimates and their confidence intervals to odds and odds ratios, and discuss their interpretation with your colleagues.

Question 3

We might prefer R to do the exponentiation process to convert to odds and odds ratios for us. Load (and install first if needed) the `broom` package. Examine the documentation for `tidy`. Then call `tidy` with your fitted logistic regression model, with suitable options to get odds and odds ratios and confidence intervals. Check your results by comparing with the ones you calculated by hand in the previous question.

Question 4

Using pen and paper and a calculator, calculate the probability that a women who weighs 120lbs (about 54.4kg) has a baby with birth weight less than 2500g.

Question 5

Load (and install if necessary first) the `marginalEffects` package. Then use the `predictions` function with your earlier fitted model to check your answer to the previous question.

Question 6

Fit an appropriate model (using logistic or logit), and perform an appropriate test, to examine whether race has an association with the probability of having a low birthweight baby, conditional on (after adjusting for) mother's weight.

What do you conclude?

Part B: Investigation into effect of CS_2 at different doses on insect survival

The data are in `insecticide.RData`. There is one record per insect with `id` indexing the insect, `dosegroup` indicating which of eight dose groups the insect belonged to, `dose` giving the actual dose of insecticide received (mg/l) in the group and `y` indicating whether or not the insect died (0/1 equating to survived/died).

The aim of the analysis is to investigate the nature of the association between dose level and the proportion of insects killed.

Question 1

Load and explore the data. Are there any missing values? How many insects do you have data for?

Question 2

Fit a logistic regression model relating whether or not the insect died to `dose` treating this as a continuous variable. Is there evidence of an increasing risk of death with increasing dose?

Question 3

Interpret the dose parameter in terms of an odds-ratio, and obtain its 95% CI.

Question 4

Use the fitted model to (manually) calculate the estimated probability of death at a dose of 55 mg/l CS_2 .

Question 5

Work out the dose that, on the basis of this fitted model, would lead to 50% of the insects dying (sometimes this is termed the LD50).

Question 6

Fit a logistic model that as well as `dose`, adds $I(\text{dose}^2)$ as a covariate. Is there evidence that adding the quadratic effect of dose improves the fit? Finally, produce a plot of the fitted values from this new model, and also the fitted values from the earlier (linear model).

Question 7

Now consider dose as a categorical/factor variable. Fit a further logistic model that treats `dose` as a factor, by using `factor(dose)` as a covariate.

Is the earlier model where we assumed a linear effect of dose (on the log odds of death) nested within the model you have just fitted?

Question 8

Carry out an appropriate likelihood ratio test to compare the fits of the models where dose is considered as continuous and categorical. What do you conclude?